

EE6008 GUIDED READING EDITION

Covert Communications by Encoding UAV Motion States: Joint Design of Codebook and Controller

Original paper interleaved with English section-by-section guidance

BIG PICTURE. The paper quantises simulated six-degree-of-freedom UAV states into a structured 16-bit label and stabilises the vehicle with a cascaded position-PID and attitude-PD controller.

ROLE IN YOUR PROJECT. Use it for the flight model, controller architecture, state logging, and a baseline quantiser. Your project must add the inverse payload encoder, independent visual receiver, framing, reliability metrics, and a formal H0/H1 visual-watcher evaluation.

How to use this edition

- Read original pages normally; blue commentary appears after the nearest page boundary for each major section.
- Use Purpose and the numbered process first, then revisit equations and figures with the vocabulary table.
- Treat red Critical Reading boxes as assumptions or evidence gaps to verify; use transfer notes for your EE6008 plan.

PLACEMENT NOTE. IEEE sections often share a source page: one section may end and the next may begin on the same page. Guide pages are therefore inserted after the closest complete source page and identify the exact section they explain.

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Covert Communications by Encoding UAV Motion States: Joint Design of Codebook and Controller

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Abstract—In this paper, we investigate the information piggyback capability of the unmanned aerial vehicle (UAV) by encoding observed motion states. Specifically, at specific moments throughout the holistic navigation process, the distance between the UAV and the starting point, the position expressed by the three-dimensional (3D) Cartesian coordinates, the linear velocities, and the attitude angles are encoded into 16-bit digital symbols through a proposed codebook. In this way, covert data communications can be enabled, complementing conventional radio frequency (RF) communications in harsh electromagnetic environments. To achieve a well-designed flight controller that is necessary to enable fluent movement against external disturbances and accurate motion state encoding, we introduce the whole flight control structure and perform system dynamic analysis. The proposed motion control mechanism can mitigate the jitter and oscillation during the journey while ensuring the required motion status for information encoding purposes.

Index Terms—Unmanned aerial vehicle (UAV), covert communications, codebook design, whole flight control.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are able to adjust their positions and movements to fit different circumstances under fully controllable three-dimensional (3D) maneuverability, which plays an important role in emergency communications requiring flexible and easy deployment [1]–[3]. Among the existing literature on UAV communications, the codebook design is a vital task for realizing efficient and reliable wireless UAV communication networks, considering the stringent requirements of beam tracking and channel estimation [4]–[8]. However, researchers mainly focus on designing the beam-forming pattern based codebook relying on radio frequency (RF) signals, but pay little attention to the UAV motion coding.

In light of this, we aim to realize covert communications by encoding UAV motion states via a predefined codebook. The proposed design brings in several advantages and can be applied in various harsh scenarios where conventional RF communications cannot work [9]. First, as the real-time status of the UAV will be encoded into digital signals and then transmitted to a receiver, the conveyed motion information facilitates tracking, channel estimation, and information exchange. Moreover, as digital symbols are encoded based on the movement and position information, UAVs can deliver additional information by following a predefined trajectory

mapping to information, which can be regarded as covert and is thus highly secured since eavesdroppers are not aware of the predefined codebook of UAV motion states.

To implement this covert communication paradigm, a precise UAV flight control system on motion states is thus imperative. In this regard, we develop an inner-outer loop based flight controller in this paper, which can accurately describe a flying system's dynamics. The linear control mechanism adopting the proportional-integral-derivative (PID) algorithm is deployed at the outer loop for the position control, and the proportional-derivative (PD) algorithm is applied in the inner loop for the attitude control. Both jointly stabilize the motion states of a UAV subjected to aerodynamic disturbances when conveying motion encoded information.

II. MODEL AND ANALYSIS OF SYSTEM DYNAMICS

A quadrotor mode with four propellers is a typical multirotor UAV, whose model can be pictorially illustrated in Fig. 1. It deploys two different sets of counter-rotating rotors at the extremities of a crossing frame. The quadrotor UAV model considered in the following assumes a symmetrical shape and a uniformly distributed body mass. In addition, gyroscopic and aerodynamic effects are omitted in this paper for simplicity. Therefore, six degrees of freedom (DoFs) of the quadrotor UAV can be manipulated, including three linear translations along the x_e , y_e , and z_e axes and three rotary translations around the three axes (roll, pitch, and yaw) in the inertial frame. Accordingly, we utilize six variables x, y, z, φ, θ , and ψ to characterize the motion variations of the six DoFs, respectively. Then, taking the UAV's center of mass as the coordinate origin O_b , three coordinate axes x_b , y_b , and z_b are drawn along the quadcopter arms in the body frame. The rotation matrix from the inertial frame to the body frame is given in (1), which is the multiplication of three canonical rotation matrices $\mathbf{C}(\varphi)$, $\mathbf{C}(\theta)$, and $\mathbf{C}(\psi)$.

The motion of the quadcopter is realized by the rotation of four rotors. Denoting the rotation speed of rotor i , $\forall i \in \{1, 2, 3, 4\}$, as ϖ_i , force F_i created in the direction of the rotor axis can be written as $F_i = K_f \varpi_i^2$, where K_f is the lift constant. Hence, thrust F_T in the z -axis can be determined by summing the forces created on each rotor as $F_T = \sum_{i=1}^4 F_i$.

I. INTRODUCTION | INSERTED AFTER SOURCE PAGE 2

The proposed communication idea and its control requirement

PURPOSE. This section proposes using sampled UAV motion states as digital symbols and argues that accurate flight control is necessary if those symbols are to be stable and interpretable.

What this section is doing

- 1 Start from the UAV's ability to move in three dimensions and note that most existing codebooks concern RF beamforming rather than physical motion.
- 2 Define a pre-shared mapping from distance, position, velocity, and attitude conditions to a 16-bit word.
- 3 Claim that a receiver can interpret those words while an eavesdropper without the table cannot understand the content.
- 4 Introduce a cascaded controller: outer-loop PID for position and inner-loop PD for attitude.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Piggyback communication	Carrying additional information on behaviour already needed for another task, here flight motion.
Codebook	A rule that maps physical states or symbols to bit patterns and, in a complete transmitter, back to feasible actions.
Motion state	A snapshot of position, velocity, attitude, or related flight variables.
Confidentiality	An unauthorised observer cannot understand message content.
Covertneess / LPD	An unauthorised observer has difficulty deciding whether communication is occurring at all.
Cascaded control	A slower outer controller sends commands to a faster inner controller.
PID / PD	Feedback controllers using proportional, integral, and derivative error terms; PD omits the integral term.

How to read the equations, figures, or evidence

- The paper's implemented direction is observed state -> 16-bit label. Look for, but do not find, an arbitrary payload input, inverse codebook, receiver observation model, or recovered payload output.
- The unknown-codebook argument concerns interpretation of content. A camera may still detect unusual motion even when it cannot decode the bits.

What transfers to the EE6008 project

- Use the paper's state quantiser and cascaded controller as a baseline.
- Define your project explicitly as payload -> feasible motion -> independent visual observation -> recovered payload.
- Add a warden named Willie and distinguish H0 normal flight from H1 the same mission with embedded payload.

CRITICAL READING. Secret mapping alone is not formal covert communication. The paper does not report false-alarm probability, missed-detection probability, ROC-AUC, detection error probability, KL divergence, or total variation distance.

Check your understanding

- What is the difference between hiding message meaning and hiding the existence of communication?
- Which block is missing if you want arbitrary payload transmission?

$$\mathbf{C}_{UAV} = \mathbf{C}(\varphi)\mathbf{C}(\theta)\mathbf{C}(\psi) = \begin{pmatrix} \cos \theta \cos \psi & \cos \theta \sin \psi & -\sin \theta \\ \sin \theta \sin \varphi \cos \psi - \cos \varphi \sin \psi & \sin \theta \sin \varphi \sin \psi + \cos \varphi \cos \psi & \sin \varphi \cos \theta \\ \sin \theta \cos \varphi \cos \psi + \sin \varphi \sin \psi & \sin \theta \cos \varphi \sin \psi - \sin \varphi \cos \psi & \cos \varphi \cos \theta \end{pmatrix} \quad (1)$$

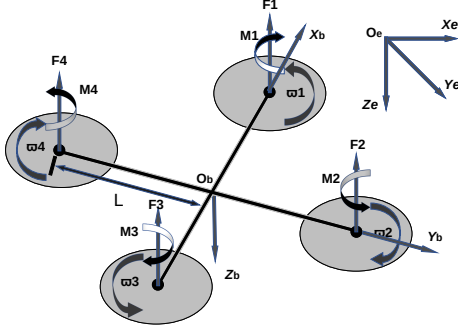


Fig. 1. Structure of the UAV coordinate system adopted in this paper.

Assuming the quadcopter to be a rigid body, Newton-Euler equations thus can be used to describe its dynamics as

$$\begin{cases} \mathbf{F}_S = m\mathbf{a} \\ \boldsymbol{\tau} = \mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} \end{cases}, \quad (2)$$

where $\mathbf{F}_S \in \mathbb{R}^{3 \times 1}$ denotes the external force added on the quadcopter measured in the inertial frame; m represents the mass of the UAV; $\mathbf{a} = [\ddot{x}, \ddot{y}, \ddot{z}]^T \in \mathbb{R}^{3 \times 1}$ is the linear acceleration of the quadcopter measured in the inertial frame; $\boldsymbol{\tau} = [\tau_x, \tau_y, \tau_z]^T \in \mathbb{R}^{3 \times 1}$ denotes the external torque; $\boldsymbol{\omega} = [\omega_\varphi, \omega_\theta, \omega_\psi]^T \in \mathbb{R}^{3 \times 1}$ and $\dot{\boldsymbol{\omega}} = [\dot{\omega}_\varphi, \dot{\omega}_\theta, \dot{\omega}_\psi]^T \in \mathbb{R}^{3 \times 1}$ are the angular velocity and acceleration vectors, respectively; $\mathbf{I} = \text{diag}\{I_x, I_y, I_z\}$ denotes the inertia matrix, which is diagonal due to the symmetric structure of the quadcopter.

In the inertial frame, the gravitational force, the drag force generated by the air resistance, and the thrust contribute to the linear acceleration of the quadcopter. The air resistance is dependent on the velocity of the motor, whose value on the inertial frame is given by $\mathbf{F}_A = [K_x \dot{x}, K_y \dot{y}, K_z \dot{z}]^T$, where K_j and $\dot{j}$, $\forall j \in \{x, y, z\}$, denote the motor rotation factor and the velocity along the j axis in the inertial frame, respectively. The gravitational force is dependent on the mass of the drone and gravity's acceleration g by $G = mg$. Therefore, denoting $\mathbf{e}_z \in \mathbb{C}^{3 \times 1}$ as the vector basis with all zeros except the third entry being one, the total external force can be calculated as $\mathbf{F}_S = \mathbf{C}_{UAV}^T F_T \mathbf{e}_z - \mathbf{F}_A - G\mathbf{e}_z$. Then, the linear acceleration of the quadcopter can be determined as $m\mathbf{a} = \mathbf{C}_{UAV}^T F_T \mathbf{e}_z - \mathbf{F}_A - G\mathbf{e}_z$, which can be expanded as

$$\begin{cases} m\ddot{x} = (\sin \theta \cos \varphi \cos \psi + \sin \varphi \sin \psi) F_T - K_x \dot{x} \\ m\ddot{y} = (\sin \theta \cos \varphi \sin \psi - \sin \varphi \cos \psi) F_T - K_y \dot{y} \\ m\ddot{z} = \cos \theta \sin \varphi F_T - K_z \dot{z} - G \end{cases}. \quad (3)$$

Besides, the analysis of the quadcopter's angular motion is also necessary as the quadcopter moves in a rotational motion around the center of body mass under the joint influence of external forces and moments. In fact, the overall external torque on the quadcopter is composed of two kinds of torques. One is the torque on each rotor resulting from



Fig. 2. Codebook structure adopted in the design for encoding motion states for a 16-bit sequence.

the angular rotation and acceleration, and the other is the gyroscopic torque, generated once the torque on the four rotors cannot cancel each other, which causes the body to deflect. Denoting M_i , $i = 1, 2, 3, 4$ as the torque generated on each rotor, it can be calculated as $M_i = K_m \omega_i^2$, where K_m denotes the drag constant. Correspondingly, the torques in the direction of the body frame can be calculated as $\boldsymbol{\tau}_R = (LK_f(-\omega_2^2 + \omega_4^2), LK_f(\omega_1^2 - \omega_3^2), K_m(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2))^T$, where L is the distance between the rotor and the center of the quadcopter's mass. Similarly, the gyroscopic torque in the body frame around each axis can be expressed as $\boldsymbol{\tau}_G = (I_r \omega_\theta \omega_\psi, -I_r \omega_\varphi \omega_\psi, 0)^T$, where $\omega_s = \omega_1 - \omega_2 + \omega_3 - \omega_4$, and I_r is the inertia moment of the rotor. Therefore, the overall external torque can be calculated as $\boldsymbol{\tau} = \boldsymbol{\tau}_R + \boldsymbol{\tau}_G$. Applying the Euler equation to the derived results above, the angular acceleration status can be calculated as $\mathbf{I}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} + \boldsymbol{\tau}_R + \boldsymbol{\tau}_G$, which can be expanded as

$$\begin{cases} I_x \dot{\omega}_\varphi = (I_y - I_z) \omega_\psi \omega_\theta + I_r \omega_\theta \omega_\psi + LK_f(-\omega_2^2 + \omega_4^2) \\ I_y \dot{\omega}_\theta = (I_z - I_x) \omega_\varphi \omega_\psi - I_r \omega_\varphi \omega_\psi + LK_f(-\omega_1^2 + \omega_3^2) \\ I_z \dot{\omega}_\psi = (I_x - I_y) \omega_\varphi \omega_\theta + K_m(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{cases}. \quad (4)$$

III. CODEBOOK DESIGN FOR COVERT COMMUNICATIONS

Based on the six DoFs, we design a codebook mapping the motion patterns throughout the flight process to digital information bits, which can be transmitted to and captured by intended receivers. As shown in Fig. 2, a novel codebook with 16-bit quantization is developed based on the distance ranges from the departure point, the orientation of the UAV phase in global coordinates (origin, quadrant, axis, and plane), the velocity direction and magnitude in each DoF of the UAV, and the rotation direction in each attitude angle of the UAV.

Recording the position vector in the inertial frame as $\mathbf{p} = [p_x, p_y, p_z] \in \mathbb{R}^{3 \times 1}$, the distance codebook is built on the distance between the UAV and the initial point. Specifically, the distance can be classified into four categories by setting three different distance thresholds, that is, $T_1 < T_2 < T_3$, without loss of generality. The distance that falls into the range from 0 to T_1 will be encoded to be '00,' while the one that falls into the range from T_1 to T_2 will be encoded to be '01.' Furthermore, when the distance is smaller than T_3 but greater than T_2 , then it will be encoded to be '10.' Otherwise, when the distance exceeds T_3 , the 2 bits will be '11.'

II. MODEL AND ANALYSIS OF SYSTEM DYNAMICS | INSERTED AFTER SOURCE PAGE 3

Turning rotor speeds into six-degree-of-freedom motion

PURPOSE. This section constructs the physical plant that the controller must command: coordinate transformations, translational forces, rotational torques, and rotor mixing.

What this section is doing

- 1 Define inertial and body coordinate frames and six states: x, y, z, roll phi, pitch theta, and yaw psi.
- 2 Use a rotation matrix to express thrust generated in the body frame as a force in the inertial frame.
- 3 Apply Newton-Euler equations: net force produces linear acceleration; net torque produces angular acceleration.
- 4 Combine gravity, drag, and rotor thrust for translation.
- 5 Combine arm-length thrust moments, rotor drag torque, inertial coupling, and gyroscopic terms for rotation.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Inertial frame	A fixed world coordinate system used to describe position and mission trajectory.
Body frame	A coordinate system attached to the UAV; its axes rotate with the vehicle.
6DoF	Six degrees of freedom: three translations plus roll, pitch, and yaw.
Rotation matrix	A matrix that converts a vector between coordinate frames without changing its physical meaning.
Rigid-body model	An approximation that ignores structural deformation of the airframe.
Inertia matrix	The rotational equivalent of mass, describing resistance to angular acceleration.
Kf / Km	Coefficients converting squared rotor speed into thrust force and reaction torque.
Gyroscopic torque	Torque produced by coupling between spinning rotors and vehicle rotation.

How to read the equations, figures, or evidence

- Eq. (1) is a coordinate transformation. Check the stated direction of the transform before multiplying a force vector.
- Eq. (2) is the compact Newton-Euler model. The cross-product term represents rotational-axis coupling.
- Eq. (3) expands translational acceleration into x, y, and z components. Verify signs and trigonometric terms against a trusted 6DoF model before implementation.
- Eq. (4) shows how rotor-speed differences create roll, pitch, and yaw moments.

What transfers to the EE6008 project

- Use a verified MathWorks 6DOF block, PX4/Gazebo model, or tested Python quadrotor model as the plant reference.
- Log both commanded and realised state; the difference determines whether a motion symbol is physically distinguishable.
- Add actuator saturation, maximum tilt, velocity, acceleration, and geofence constraints before hardware flight.

CRITICAL READING. The text first says aerodynamic and gyroscopic effects are omitted, then introduces drag and gyroscopic torque. Some expanded equations also appear dimensionally or trigonometrically inconsistent. Treat the section as an architecture guide and validate every equation independently.

Check your understanding

- Why must thrust be rotated from the body frame into the inertial frame?
- Which rotor-speed combinations create roll, pitch, yaw, and total thrust?

In addition, we adopt four bits from b_3 to b_6 to encode 15 possible positions of the UAV. The first bit b_3 classifies the position of the UAV into two types. One category is that the UAV is located at one of the eight quadrants rather than on the coordinate axes or the large planes¹, that is, $b_3 = 0$ when $p_x \neq 0$, $p_y \neq 0$, and $p_z \neq 0$. Otherwise, $b_3 = 1$ indicates the scenario where the UAV is on one of the three coordinate axes or one of the three large planes, that is, $p_x = 0$ or $p_y = 0$ or $p_z = 0$. The fourth bit denoted as b_4 is determined by the position of the UAV in terms of x_e -axis. When the UAV is located in one of the eight quadrants, that is, $b_3 = 0$, we set $b_4 = 1$ when $p_x > 0$ and $b_4 = 0$ when $p_x < 0$. Otherwise, when $b_3 = 1$, we set $b_4 = 1$ when $p_x = 0$ and $b_4 = 0$ when $p_x \neq 0$. Also, we can obtain b_5 and b_6 in the same way as setting b_4 , but with respect to the y_e -axis and z_e -axis, respectively. The position codebook is summarized in Table I.

Subsequently, we target encoding the linear velocities into digital information bits. We assign b_7 to encode the magnitude of the overall velocity, b_8 and b_9 to encode the velocity along the x_e -axis, b_{10} and b_{11} to encode the velocity along the y_e -axis, and the remaining two bits, i.e., b_{12} and b_{13} , to encode the velocity along the z_e -axis. We can do so by first determining the scalar magnitude of the UAV ensemble velocity via $\|\mathbf{v}\| = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$, and then comparing it with a predefined velocity threshold v_T . If $\|\mathbf{v}\| \geq v_T$, then $b_7 = 1$, or $b_7 = 0$, otherwise. Then, the velocity direction along the x_e -axis can be reflected by the value of b_8 when comparing $\dot{x}$ with 0. Without loss of generality, b_8 is set to be 1 when $\dot{x} > 0$ or be 0, otherwise. The magnitude of $\dot{x}$ is then compared with the velocity threshold $v_{T,x}$ to ensure the quadcopter moving along x_e -axis at a desired speed, where $b_9 = 1$ when $|\dot{x}| \geq v_{T,x}$ and $b_9 = 0$ when $|\dot{x}| < v_{T,x}$. Similarly, bits from b_{10} to b_{13} can be obtained in the same way by comparing $\dot{y}$ and $\dot{z}$ with 0 and their threshold $v_{T,y}$, $v_{T,z}$, respectively. In this way, the real-time direction and magnitude of the velocity in each of the three axes in the 3D coordinate system can be encoded. The codebook of the linear velocity is summarized in Table II.

In addition to the linear velocity, the rotation variation of the UAV is also important information that is normally required to be recorded by most standards, especially when the trajectory is not predefined but dynamically adjusted by the controller. Moreover, the rotation status can also be encoded to carry additional information. Therefore, we develop a 3-bit codebook to record the instantaneous rotation direction of the three Euler angles φ , θ , ψ . The corresponding information bit is set to be 1 if the rotation direction is non-negative, or 0, otherwise. For instance, the direction information of angle φ rotating around the x_e -axis is encoded in b_{14} , which is set to be 1 if $\varphi \geq 0$, or be 0 if $\varphi < 0$. Likewise, the direction information of angles θ and ψ is encoded in b_{15} and b_{16} , respectively, in a similar way. The codebook for encoding the attitude angles is summarized in Table III.

¹In order to access the relative position of the UAV with respect to the starting point, we set up a 3D coordinate by letting the starting point as the origin. There are 8 quadrants, 3 coordinate axes, and 3 large planes.

Algorithm 1 Holistic encoding process of the 16-bit sequence.

- 1: **Input:** Position of UAV $\mathbf{p}$, three distance threshold $T_1 < T_2 < T_3$, linear velocity vector $\mathbf{v}$, velocity thresholds v_T , $v_{T,x}$, $v_{T,y}$, $v_{T,z}$, and attitude angles φ , θ , ψ .
- 2: Utilize position information $\mathbf{p}$ and three distance threshold $T_1 < T_2 < T_3$ to yield b_1 and b_2 ;
- 3: Utilize position information $\mathbf{p}$ to yield bits from b_3 to b_6 based on the position codebook developed in Table I;
- 4: Utilize linear velocity vector $\mathbf{v}$ and velocity thresholds v_T , $v_{T,x}$, $v_{T,y}$, and $v_{T,z}$ to yield bits from b_7 to b_{13} based on the linear velocity codebook developed in Table II;
- 5: Utilize the attitude angles φ , θ , and ψ to yield bits from b_{14} to b_{16} based on the codebook developed in Table III;
- 6: **Output:** 16-bit sequence, i.e., $[b_1, b_2, \dots, b_{16}]$.

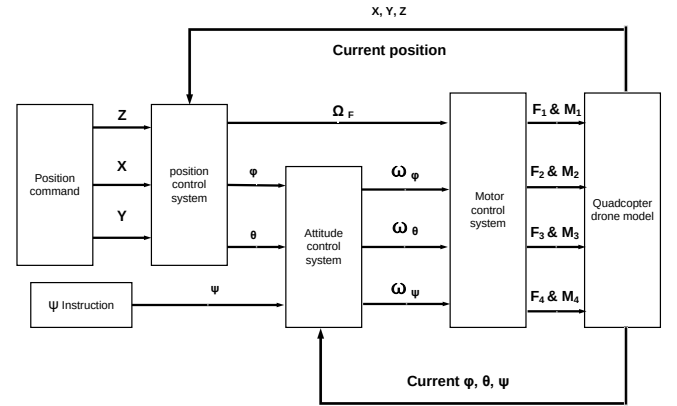


Fig. 3. Quadrotor UAV control structure.

Assembling the four developed codebooks above from the perspectives of distance, position, linear velocity, and attitude angles, the 16-bit codebook as a whole for encoding the UAV's motion information can be established. The overall 16-bit sequence encoding process is summarized in Algorithm 1.

IV. INTER-OUTER LOOP BASED CONTROLLER DESIGN

As the codebook designed in the last section is feasible when the quadcopter can be operated precisely as per the instructions from the flight controller, we propose the whole flight control structure based on the motion dynamics analyzed in Section II, aiming to lay the foundation for both simulations and practical implementations. Adopting inner and outer control loops for coordinating across subsystems, Fig. 3 shows the implemented controller architecture on the quadcopter system, which is composed of position control, attitude control, quadcopter dynamics system, and motor dynamics subsystem as discussed in the sequel.

The position control module leverages the desired position information (x_d, y_d, z_d) and the feedback of the instantaneous actual position information (x_t, y_t, z_t) to regulate the on-board altitude and calculate the required roll and pitch angles. Moreover, according to the quadrotor UAV hovering in the air, the propeller generates a pulling force equal to the gravity of

III. CODEBOOK DESIGN FOR COVERT COMMUNICATIONS | INSERTED AFTER SOURCE PAGE 4

How a flight-state snapshot becomes 16 bits

PURPOSE. This section quantises a continuous UAV state into a structured digital label. It is a forward state encoder, not yet a transmitter that chooses motion from payload bits.

What this section is doing

- 1 Use bits b1-b2 for four distance bands defined by thresholds $T1 < T2 < T3$.
- 2 Use b3-b6 for fifteen position classes: eight octants plus planes, axes, and the origin logic.
- 3 Use b7 for total-speed threshold and b8-b13 for each axis's direction and magnitude threshold.
- 4 Use b14-b16 for the sign of roll, pitch, and yaw angles.
- 5 Pack the fields in Algorithm 1 to produce one 16-bit label per sampled state.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Quantisation	Converting a continuous measurement into one of a finite number of categories.
Codeword	The resulting digital label, here a structured 16-bit sequence.
Octant	One of eight 3D regions defined by the signs of x, y, and z.
Threshold	A boundary that assigns a measured value to one discrete category or another.
Guard band	An unused region around a threshold that reduces ambiguous switching under noise.
Hysteresis	Using different switch-up and switch-down thresholds so a noisy value does not chatter.
Injective mapping	A one-to-one map where different inputs always have different outputs.
Feasible codeword	A bit pattern consistent with the coupled physical constraints of the UAV state.

How to read the equations, figures, or evidence

- Bits are not independent. Distance depends on position; total speed depends on axis speeds; many nominal 16-bit combinations cannot occur physically.
- Continuous states are grouped into the same label, so the map is many-to-one from physical state to codeword, despite the paper's later use of the term injective.
- The attitude field is described as rotation direction, but the rule tests angle sign, not angular-velocity sign.
- Exact equality to $x=0$, $y=0$, or $z=0$ is fragile in real sensors. Use tolerances and hysteresis.

What transfers to the EE6008 project

- Reproduce the bit packing as a simulation baseline and unit-test every boundary condition.
- Create the missing inverse encoder: payload symbol $\rightarrow$ dynamically reachable target state or trajectory primitive.
- Use relative states around a nominal path rather than absolute global octants for a small indoor course project.
- Protect thresholds with guard bands and reject uncertain observations instead of forcing a bit decision.

CRITICAL READING. The distance and velocity threshold values are not published. The codeword space is constrained, and sampling every 20 ms does not imply a 16-bit message every 20 ms or an 800 bit/s channel.

Check your understanding

- Can every one of the 65,536 nominal bit patterns be realised? Why not?
- What must be added to convert this state logger into a payload transmitter?

Table I. Position mapping relation for encoding based on the 3D inertial frame.

Value \ bits	b_3	b_4	b_5	b_6
1	$p_x \parallel p_y \parallel p_z = 0$	$(b_3 = 1 \ \& \ p_x = 0)$ or $(b_3 = 0 \ \& \ p_x > 0)$	$(b_3 = 1 \ \& \ p_y = 0)$ or $(b_3 = 0 \ \& \ p_y > 0)$	$(b_3 = 1 \ \& \ p_z = 0)$ or $(b_3 = 0 \ \& \ p_z > 0)$
0	$p_x \ \& \ p_y \ \& \ p_z \neq 0$	$(b_3 = 1 \ \& \ p_x \neq 0)$ or $(b_3 = 0 \ \& \ p_x < 0)$	$(b_3 = 1 \ \& \ p_y \neq 0)$ or $(b_3 = 0 \ \& \ p_y < 0)$	$(b_3 = 1 \ \& \ p_z \neq 0)$ or $(b_3 = 0 \ \& \ p_z < 0)$

Table II. Linear velocity mapping relation for encoding.

Value \ bits	b_7	b_8	b_9	b_{10}	b_{11}	b_{12}	b_{13}
1	$ \mathbf{v} \geq v_T$	$\dot{x} \geq 0$	$ \dot{x} \geq v_{T,x}$	$\dot{y} \geq 0$	$ \dot{y} \geq v_{T,y}$	$\dot{z} \geq 0$	$ \dot{z} \geq v_{T,z}$
0	$ \mathbf{v} < v_T$	$\dot{x} < 0$	$ \dot{x} < v_{T,x}$	$\dot{y} < 0$	$ \dot{y} < v_{T,y}$	$\dot{z} < 0$	$ \dot{z} < v_{T,z}$

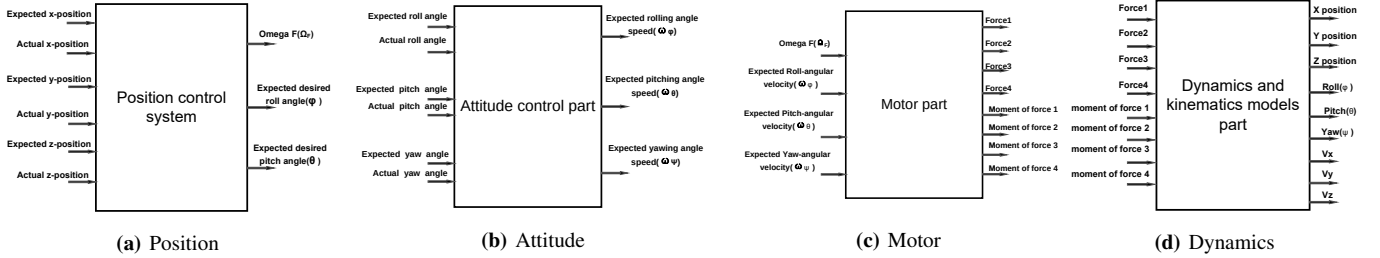


Fig. 4. Control modules and their input-output relations.

Table III. Attitude angle mapping relation for encoding.

Value \ bits	b_{14}	b_{15}	b_{16}
1	$\varphi \geq 0,$	$\theta \geq 0$	$\psi \geq 0$
0	$\varphi < 0,$	$\theta < 0$	$\psi < 0$

the quadrotor in quantity. Based on this relation, the hovering speed of the quadrotor Ω_h can be calculated by using the propeller pull factor K_f as $\Omega_h = \sqrt{mg/(4K_f)}$.

Therefore, as shown in Fig. 4a, the input of the position control is the desired and actual position information, while its outputs are the total thrust in the vertical direction Ω_F affecting the acceleration in the z -direction and the roll and pitch angles. According to (3), the thrust on the quadcopter depends on its linear acceleration, which can be expressed as

$$\Omega_F = (\ddot{z} + e_z) \sqrt{m/(gK_f)} + \sqrt{mg/(4K_f)}. \quad (5)$$

Roll and pitch angles, i.e., ϕ and θ , are also related to the linear accelerations, which can be derived as

$$\begin{cases} \phi = -g^{-1} [(\ddot{x} + e_x) \sin(\psi) - (\ddot{y} + e_y) \cos(\psi)] \\ \theta = -g^{-1} [(\ddot{x} + e_x) \cos(\psi) + (\ddot{y} + e_y) \sin(\psi)] \end{cases}, \quad (6)$$

where $e_i = i_d - i_t$, $i \in \{x, y, z\}$ is the error signal, and $\ddot{i}$ is the stabilized acceleration output of the controller.

The PID controller is introduced in this module to stabilize the hover position and the attitude. Based on the input desired and measured position states, the position errors along each direction can thus be computed by their difference, which

can be utilized by the PID controller to regulate the linear accelerations through

$$\ddot{i} = K_{P_i} e_i + K_{I_i} \int e_i dt + K_{D_i} \dot{e}_i, \quad (7)$$

where K_{P_i} , K_{I_i} , and K_{D_i} are the proportional, integral and derivative gains for the PID controller. These three parameters can be orchestrated to maintain stable motion states subjected to aerodynamic jitters. Then, bringing the well-controlled linear accelerations from (7) back to (5) and (6), more stable thrust and attitude can be achieved.

For controlling the attitude of the quadcopter, namely, the rolling speeds of the roll, pitch, and yaw angles, we employ a PD controller, with which a simpler control structure design gives a quicker response² [10]. Accordingly, with desired attitude angles φ_d , θ_d , and ψ_d , and the feedback of actual angles φ_t , θ_t , and ψ_t , the speed of these attitude angles can be calculated as $\omega_j = K_{P_j} e_j + K_{D_j} \dot{e}_j$, $\forall j \in \{\varphi, \theta, \psi\}$, where K_{P_j} is the proportional regulation acting as the mainstay; K_{D_j} is the differential regulation working as a supplement; $e_j = j_d - j_t$ denoting the difference between the desired and actually measured angles. The controllable regulation parameters can also be adjusted during the flying process to stabilize the angular motion as shown in Fig. 4b. The obtained angular velocities are utilized in the motor dynamics system to design the rotation speed of the UAV motor.

With the produced UAV's three angular velocities and vertical propeller thrust Ω_F , a motor dynamic module is designed and implemented to convert them into the rotation

²Timeliness is crucial since the inner loop should be faster than the outer loop to avoid cross-loop synchronization issues.

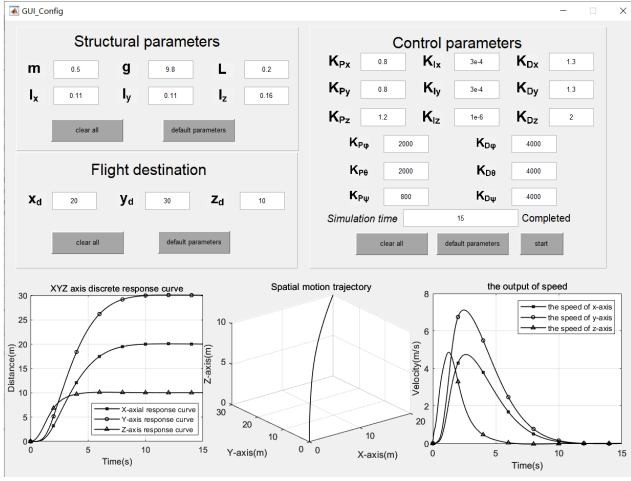


Fig. 5. Simulation GUI developed on MATLAB and the results generated by the given parameter set.

speed of the UAV's four rotors, which can be subsequently translated into the external forces and the torque of the four rotors. The structure of the motor model is shown in Fig. 4c, which treats the angular velocity output of the attitude control as the input and takes the forces as well as the moments of the forces on each rotor as the output. The driving force that can be expressed as a function of the rotor speed as the power supplied to the system by the motor is the sum of the squares of the rotor speeds, while the UAV's three angular velocities are also related to the rotor speeds according to (4). These interconnected relations form a dynamic system that can be expressed in the form of an equation set as

$$\begin{cases} \Omega_F = K_f (\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\ \omega_\phi = LK_f (-\omega_2^2 + \omega_4^2) \\ \omega_\theta = LK_f (\omega_1^2 - \omega_3^2) \\ \omega_\psi = K_m (\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{cases} \quad (8)$$

From the above, we can obtain the speed control quantities ω_i^2 , $\forall i \in \{1, 2, 3, 4\}$ for each of the four motors, which can be utilized to derive the forces and moments in the four motors of the UAV through $F_i = K_f \omega_i^2$ and $M_i = K_m \omega_i^2$.

The module shown in Fig. 4d presents the dynamics and kinematics of the UAV resulting from the generated force from the motor dynamics system. In principle, by inputting the force information, the module can figure out the measured position, the Euler angles, and the linear velocities³.

V. PROOF OF CONCEPT BY SIMULATION RESULTS

In this section, we utilize the Simulink toolbox on MATLAB R2022b to develop the overall flight controller based on the analysis in Section IV. The adopted simulation parameters

³For the software implementation, the 6DOF module under MATLAB R2022b allows for the conversion of the input forces and moments on the UAV into desired velocities, positions, and Euler angles in the geodetic coordinate system. As this is out of the scope of this paper, more details of the 6DOF module can be found in [11], which are not repeated in this paper for brevity.

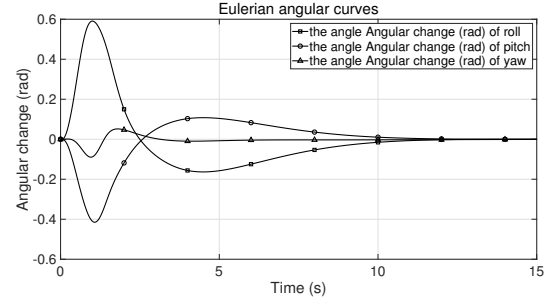


Fig. 6. Euler angle output over time.

Table IV. Parameter set adopted for the simulations in this paper.

Parameter (notation)	Value
Mass (m)	0.5 kg
Gravitational acceleration (g)	9.8 m/s ²
x/y/z-axis moment of inertia ($I_{\{.\}}$)	0.11/0.11/0.16 kg·m ²
UAV arm length (L)	0.2 m
Yaw (ψ)	0 rad
Propeller pull factor (K_f)	6.1×10^{-8}
Propeller torque factor (K_m)	1.5×10^{-9}
x/y/z-axis proportionality factor ($K_{P\{.\}}$)	0.8/0.8/1.2
x/y/z-axis integral factor ($K_{I\{.\}}$)	100/300/1 $\times 10^{-6}$
x/y/z-axis differential factor ($K_{D\{.\}}$)	1.3/1.3/2.0
$\phi/\theta/\psi$ proportionality factor ($K_{P\{.\}}$)	2,000/2,000/800
$\phi/\theta/\psi$ differential factor ($K_{D\{.\}}$)	4,000/4,000/4,000

are given in Table IV. To facilitate the simulation, parameter adjustment, and flying pattern observations, a graphical user interface (GUI) is developed as shown in Fig. 5. The developed GUI is composed of four parts, including the structure parameters, the flight destination, the control parameters, and the image output. As shown in Fig. 6, we set the yaw angle to 0 degree so that the UAV has an Euler angle of 0 at the initial position. The UAV can be rapidly stabilized by adapting the three Euler angles after departure and finally reaches the target position. The change in the yaw angle reflects that the fuselage jitters of the UAV throughout the entire journey are trivial. The overall controller design is illustrated in Fig. 7.

With the generated motion patterns, the digital information bits can be produced based on the codebook designed in Section III. Once the UAV starts flying, the digital data sequences will continuously be generated throughout the whole flight. The output digital data sequences are shown in Fig. 8, which provide useful insights into the instantaneous flight motion states. Each line of the bit stream is the result of the motion state encoded during the flight of the UAV, thus being able to reflect the motion information of the UAV. Taking the fourth data stream as an example, we can acquire from the first two bits that the UAV is located within ten meters from the origin. Then, the third to the sixth bit, that is, '0110,' indicates that the UAV is located at the fifth quadrant of the inertial frame with $x > 0$, $y > 0$, and $z < 0$. Also, the linear velocity bits reflect the overall linear velocity magnitude, and the one along each axis does not exceed the thresholds. However, the UAV is flying along the positive direction of each axis. The last three bits imply that the UAV rotates by a positive roll

Making desired motion symbols physically reachable

PURPOSE. This section closes the feedback loop. The outer controller converts position error into thrust and desired attitude; the inner controller makes the UAV track that attitude through rotor commands.

What this section is doing

- 1 Subtract measured position from desired position to obtain x, y, and z tracking errors.
- 2 Apply outer-loop PID to generate stabilised linear-acceleration commands.
- 3 Convert the acceleration commands into total thrust and desired roll/pitch while yaw is handled separately.
- 4 Apply a faster inner-loop PD controller to attitude error and produce roll, pitch, and yaw rate or moment commands.
- 5 Use the motor mixer to convert total thrust and angular commands into four rotor-speed or force commands.
- 6 Feed the resulting simulated position, velocity, and Euler angles back to the controllers.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Setpoint	The desired state that a feedback controller is asked to track.
Feedback	Measured output returned to the controller so it can correct error.
Proportional term	Correction proportional to current error; high gain reacts strongly but may oscillate.
Integral term	Correction based on accumulated error; removes bias but can wind up under saturation.
Derivative term	Correction based on error change; adds damping but is sensitive to noise.
Settling time	Time required for the response to enter and remain near the target.
Overshoot	How far the response passes beyond the target.
Motor mixer	The allocation rule from total thrust and body moments to individual rotor commands.
Anti-windup	A mechanism that prevents the integral term from growing while actuators are saturated.

How to read the equations, figures, or evidence

- Fig. 3 is the control-loop overview. Fig. 4 decomposes it into position, attitude, motor, and dynamics modules.
- Eqs. (5)-(7) express the outer-loop calculation: position error -> acceleration -> thrust and desired attitude.
- The inner loop must be faster than the outer loop; otherwise the outer loop commands a plant whose attitude has not settled.
- Eq. (8) is intended as a mixer relation, but several left-hand quantities are labelled angular velocities where torques or moments would normally be expected. Check dimensions before copying it.

What transfers to the EE6008 project

- Command high-level bounded setpoints through an established Crazyflie interface instead of replacing low-level firmware in the MVP.

- Use jerk-limited or minimum-jerk smoothing between motion symbols so the codebook does not demand discontinuous acceleration.
- Measure tracking RMSE, overshoot, settling time, jerk, tilt, and actuator saturation for every symbol.
- Add failsafe states: estimator loss, geofence breach, excessive tilt, low battery, and immediate return to hover/land.

CRITICAL READING. The paper does not specify solver configuration, actuator limits, maximum RPM, anti-windup, noise, wind model, or all drag/inertia parameters. Stable nominal simulation is therefore not sufficient for hardware readiness.

Check your understanding

- What information crosses from the outer loop to the inner loop?
- Why should a communication symbol planner know the controller's acceleration and settling limits?

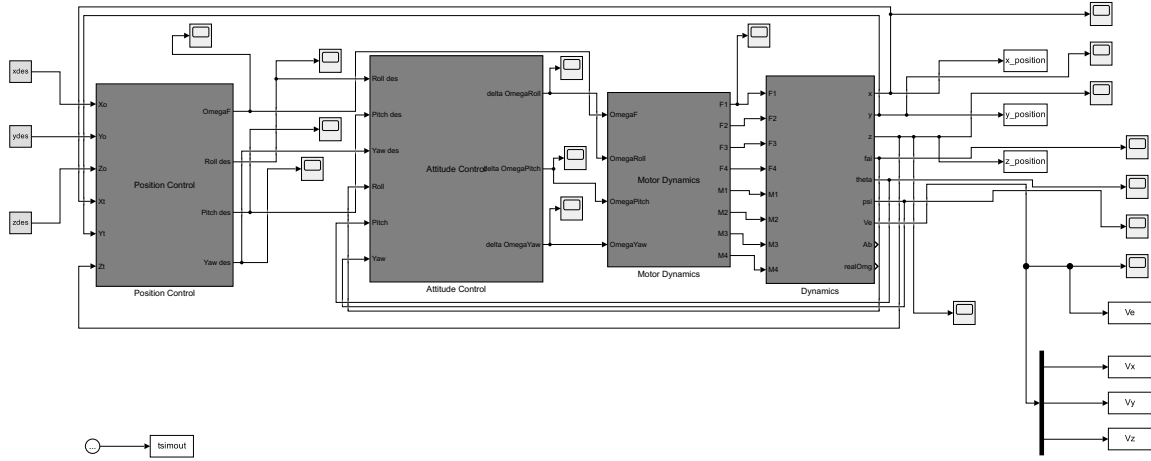


Fig. 7. Holistic flight control structure proposed in this paper and constructed on Simulink.

Data group 1	0	0	1	0	0	0	0	1	0	1	0	1	0	1	1	1
Data group 2	0	0	1	0	0	0	0	1	0	0	0	1	0	1	1	1
Data group 3	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1
Data group 4	0	0	0	1	1	0	0	1	0	1	0	1	0	1	0	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Data group 752	1	1	0	1	1	0	0	0	0	0	0	0	0	1	0	0

Fig. 8. Encoded 16-bit digital data sequence as the output within a single sample frame of 0.02 second.

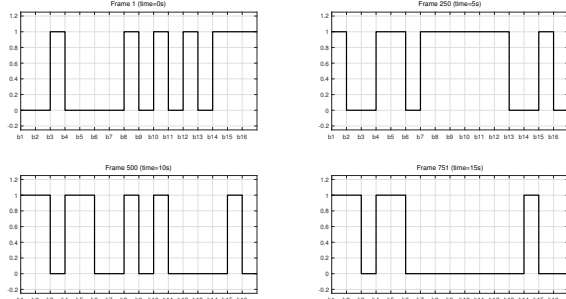


Fig. 9. Examples of digital data generated by encoding real-time movements at different frames.

angle, a negative pitch angle, and a negative yaw angle.

To summarize these simulation results, with the proposed motion state based codebook, we have successfully constructed an injective mapping relation between the real-time motions of the UAV and 16-bit data sequences. The real-time motion of the UAV is recorded and displayed in Fig. 9 as a demonstration of this injective mapping relation. In this way, the covertness and secrecy of UAV assisted wireless communications can be greatly enhanced.

VI. CONCLUSION

This work presents novel designs of UAV motion state-based codebook and flight controller for covert communications. Simulations on Simulink/MATLAB validated the feasibility of this approach, where the codebook encodes the UAV's six DoFs into 16-bit sequences, enhancing information

transmission and covertness. The flight controller, structured by inner and outer loops, can stabilize the motion of the UAV when conveying motion encoded information. Simulations also corroborated stable UAV navigation while encoding data into motion states in this way, showing an interdisciplinary area between communication and control sciences.

ACKNOWLEDGMENT

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V. PROOF OF CONCEPT BY SIMULATION RESULTS | INSERTED AFTER SOURCE PAGE 7

What the Simulink demonstration actually validates

PURPOSE. This section demonstrates stable point-to-point simulated flight and continuously labels the simulated state. It validates a state logger plus controller, not yet an independent communication link.

What this section is doing

- 1 Implement the controller in MATLAB/Simulink R2022b using the listed vehicle and controller parameters.
- 2 Simulate a 15-second flight and inspect Euler-angle transients and approach to the target position.
- 3 Sample the simulated state every 0.02 seconds, yielding 751 samples including time zero.
- 4 Apply the 16-bit state quantiser at each sample and display selected bit patterns over the flight.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Proof of concept	A limited demonstration that a core idea can run, not a complete performance validation.
Sample period	Time between state observations; 0.02 s corresponds to a 50 Hz logging rate.
Transient	The temporary response before the system reaches steady behaviour.
Ground truth	The trusted state used for evaluation; here it comes directly from the simulator.
RMSE	Root Mean Square Error, a standard magnitude measure for tracking differences.
BER / PER	Bit and packet error rates measured after a receiver attempts to recover transmitted data.
Confidence interval	A range expressing statistical uncertainty over repeated independent trials.

How to read the equations, figures, or evidence

- Fig. 6 shows attitude transients, while Figs. 8-9 show labels computed directly from simulated states. There is no external camera, state estimator, decoder error, or communication channel.
- The 50 Hz sample rate is logging resolution. Adjacent states are strongly correlated and cannot each be treated as an independent 16-bit payload.
- The paper gives vehicle mass, inertia, arm length, K_f , K_m , and controller gains, but omits the actual distance and speed thresholds required to reproduce the displayed bits.

What transfers to the EE6008 project

- First reproduce nominal state trajectories and bit packing; verify expected bit changes with automated tests.
- Next feed an arbitrary payload to an inverse motion planner and recover it from independently generated noisy visual observations.
- Evaluate at least 20 independent flights per feasible condition and report BER/PER, RMSE, overhead, and confidence intervals.
- Freeze train/test flight sessions when building Willie so the detector is evaluated on unseen flights.

CRITICAL READING. No thresholds, code, Simulink model, solver, receiver, BER/PER, energy, disturbance sweep, or formal covertness metric are reported. The result supports feasibility of state labelling and cascaded control only.

Check your understanding

- Why is state sampling not equivalent to message transmission?
- Which experiment would prove payload-to-motion-to-payload recovery?

VI. CONCLUSION | INSERTED AFTER SOURCE PAGE 7

The paper's contribution and the extension path

PURPOSE. The conclusion frames the work as an intersection of communication and control. For your project, its main value is a baseline architecture and an explicit list of missing blocks to implement.

What this section is doing

- 1 Restate the 16-bit state codebook and cascaded flight controller.
- 2 Claim stable simulated navigation while state labels are continuously generated.
- 3 Position motion-state encoding as a new interdisciplinary direction.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Interdisciplinary	Combining communication-system concepts with dynamics and control.
State telemetry	Reporting the state a vehicle already has.
Commanded communication	Choosing a physical state or trajectory because a payload symbol must be sent.
Threat model	A precise statement of the observer, its measurements, knowledge, training access, and decision task.

How to read the equations, figures, or evidence

- The strongest supported statement is: simulated flight states can be stably generated and mapped to structured labels.
- The unsupported leap is: these labels automatically create a covert, secure, high-rate communication channel.

What transfers to the EE6008 project

- State the EE6008 objective as reproducing the state-to-label baseline and adding inverse encoding, a visual receiver, protocol framing, and formal detectability evaluation.
- Ask the supervisor for the original SLX/scripts, threshold values, solver settings, errata, intended payload interpretation, and available laboratory hardware.
- Use a negative-result-friendly success criterion: a fully measured Pareto boundary is valuable even if Willie detects every reliable codebook.

CRITICAL READING. Do not present the original 16-bit label as a measured 800 bit/s covert channel. Make explicit which results are reproduced, which are extensions, and which claims remain hypotheses.

Check your understanding

- Write one sentence describing what the paper demonstrates without using the word covert.
- List the four blocks your project must add to create a complete communication system.